Retail Sales Analysis Project

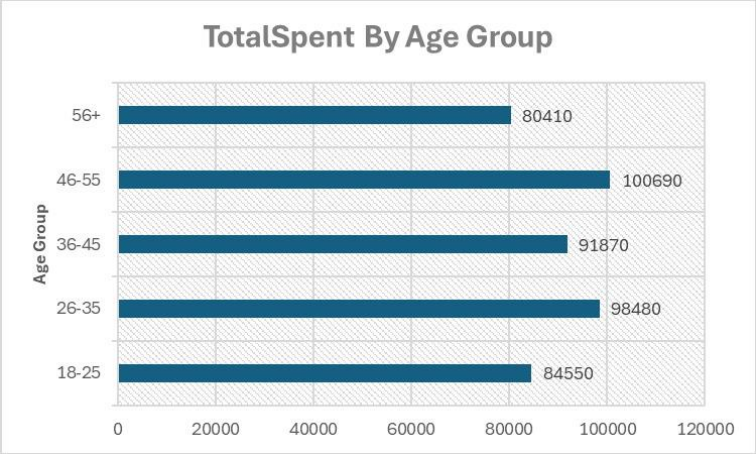
Internship Report

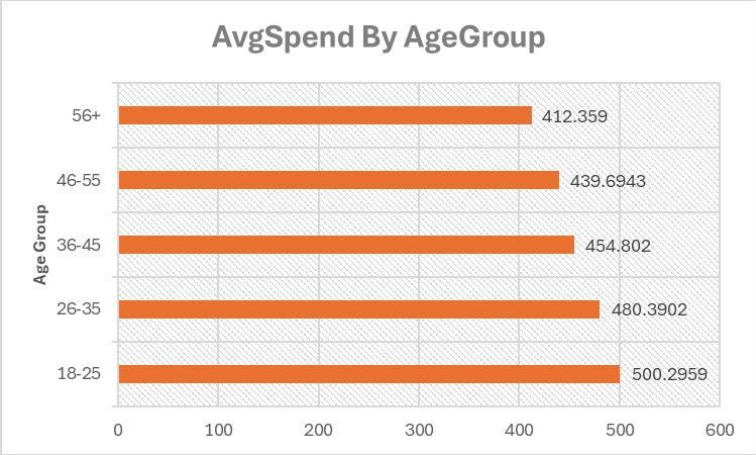
Introduction

This report presents the exploratory data analysis (EDA) of the retail sales dataset.

The objective is to uncover customer demographics, product performance, transaction behavior, and seasonal trends.

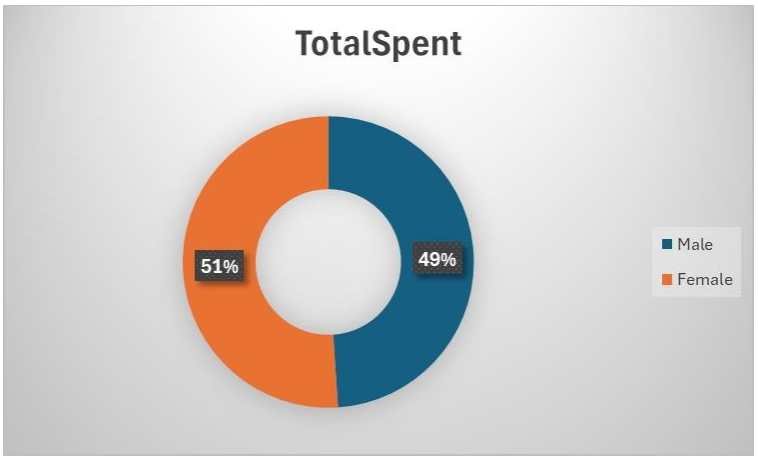
Age Group Spending (Bar Chart)

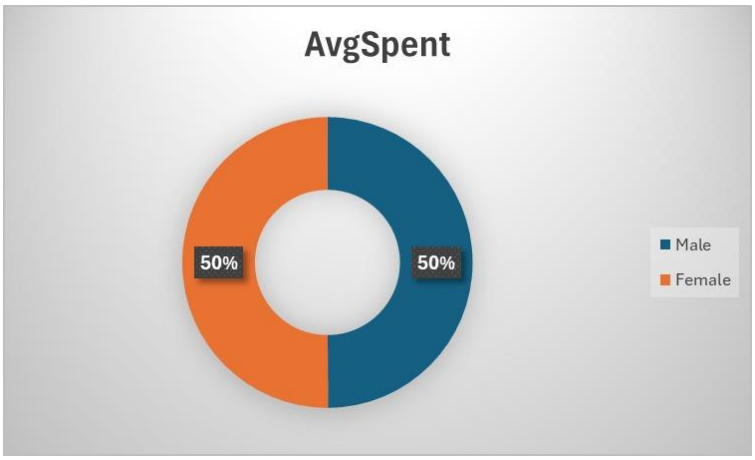




Insight: Age group 46–55 has the highest total spend, while 18–25 shows the highest average spend per transaction.

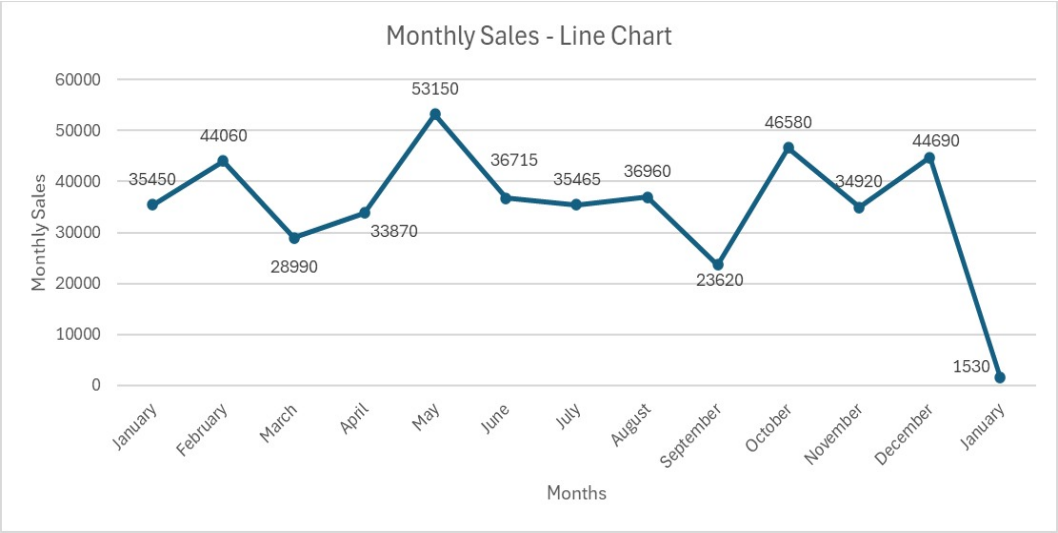
Gender Spending (Pie Chart)





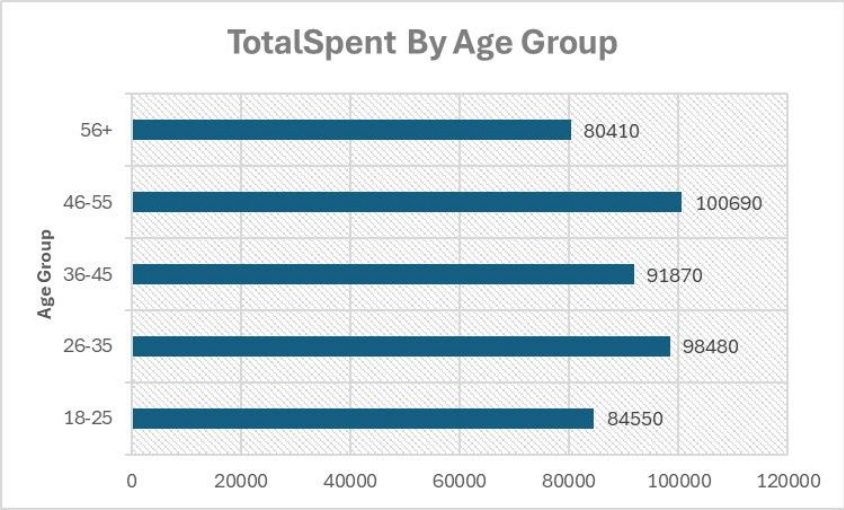
Insight: Female customers slightly outspend males, indicating potential for gender-based promotions.

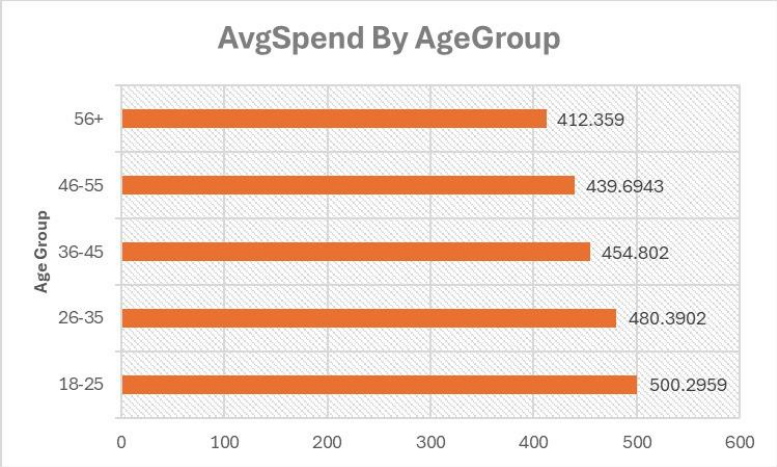
Monthly Sales Trends (Line Chart)



Insight: Peaks in May, October, and December highlight festive demand. September shows the lowest sales.

Total and Average Spending by Age Group (Bar Chart)





Insight: The 46–55 age group shows the highest total spending (₹100,690), while the 18–25 group records the highest average spend per transaction (~₹500). Older customers (56+) spend the least both in total and average terms.

Customer Frequency (Heatmap)

The customer frequency heatmap has been exported as a PDF and is available in the Visualizations folder on the main branch for reference.

Insight: Clear segmentation of premium, mid-range, and budget shoppers is visible.

Findings

- Age group 46–55 contributes the highest overall revenue, highlighting them as a key demographic.

- Younger customers (18–25) spend more per transaction, showing strong potential for premium product targeting.

- Spending gradually decreases in older age groups (56+), indicating lower engagement.

- Combining total and average spend provides a balanced view of both volume and per-transaction behavior.

Reports

- 1000 transactions analyzed across five age groups.

- Clear demographic segmentation: high total spent in middle-aged customers, high average spent in younger customers.

- Insights highlight opportunities for age-specific marketing strategies.

Conclusion

The combined analysis of total and average spending by age group reveals that middle - aged customers (46–55) drive overall revenue, while younger customers (18–25) show higher per-transaction value. This dual perspective strengthens understanding of customer behavior.

Recommendations

- Target 46–55 age group with loyalty programs to sustain high total spending.

- Offer premium product bundles to 18–25 customers who spend more per transaction.

- Design differentiated campaigns for mid-range (26–45) and budget-oriented (56+) segments.

- Use age-based segmentation in promotions to maximize both transaction volume and per-transaction value.